

Question 1: How many orders did the entire chain sell in 2021? A single number is needed for the presentation slide

Thinking Process

- Determine the counting unit: 1 order (order_number) has multiple product lines (line_item), so we must count DISTINCT order_number; using a regular COUNT will result in duplicates.
- Filter by year using YEAR(order_date) = 2021.

SQL

```
SELECT  
  COUNT(DISTINCT [order_number]) AS sum_order  
FROM [retails].[sales]  
WHERE YEAR([order_date]) = '2021'
```

Query Result

	sum_order
1	498

Insight

- In 2021, there were 498 orders, but this number CANNOT be directly compared with previous years, because the data ends on February 20, 2021, which is only about 7 weeks into the year.
- This data is purely for reporting and display purposes.
- There is not enough contextual information to make comparisons and reach accurate conclusions and decisions.

Question 2: List all categories along with the number of SKUs for each type, sorted alphabetically to create a product mix overview.

Thinking Process

- The products table contains 1 SKU per row -> use GROUP BY category and COUNT(product_key).
- ORDER BY category to sort alphabetically as requested.

SQL

```
SELECT
  [category]
, COUNT([product_key]) AS sum_category_number
FROM [retails].[products]
GROUP BY [category]
ORDER BY [category]
```

Query Result

	category	sum_category_number
1	Audio	115
2	Cameras and camcorders	372
3	Cell phones	285
4	Computers	606
5	Games and Toys	166
6	Home Appliances	661
7	Music, Movies and Audio Books	90
8	TV and Video	222

Insight

- The portfolio has 8 categories with a total of 2,517 SKUs, but the distribution is highly skewed: Home Appliances (661) and Computers (606) account for about 50% of all SKUs.
- Music, Movies and Audio Books has only 90 SKUs (about 3.6%), making it the thinnest category. We should also rely on revenue, profit, and sales velocity to decide whether to maintain or streamline this category.

Question 3: Top 10 cities with the highest number of customers (city, state, country, number of customers) to choose a location for an appreciation event.

Thinking Process

- GROUP BY 3 levels: city + state + country (to avoid mistakenly grouping 2 cities with the same name located in 2 different states).
- Use COUNT(DISTINCT customer_key), sort descending, and take TOP 10.

SQL

```
SELECT
  TOP 10
  cs.[city]
, cs.[state]
, cs.[country]
```

```
, COUNT( DISTINCT CS.[customer_key]) AS total_customer
FROM [retails].[customers] cs
GROUP BY
    cs.[city]
    , cs.[state]
    , cs.[country]
ORDER BY total_customer DESC
```

Query Result

	city	state	country	total_customer
1	Toronto	Ontario	Canada	203
2	New York	New York	United States	130
3	Los Angeles	California	United States	119
4	Montreal	Quebec	Canada	97
5	Chicago	Illinois	United States	90
6	Houston	Texas	United States	84
7	Calgary	Alberta	Canada	70
8	Dallas	Texas	United States	67
9	Vancouver	British Columbia	Canada	59
10	Philadelphia	Pennsylvania	United States	53

Insight

- Toronto leads absolutely with 203 customers, nearly 60% higher than the second place, New York (130). If only one location is chosen for the event, Toronto is a highly suitable choice.
- Interesting point: 4 out of the 10 cities belong to Canada (Toronto, Montreal, Calgary, Vancouver), even though Canada only has 5 physical stores, proving that Canadian customers mainly come from the online channel. An appreciation event here is also an opportunity to "meet" the group of customers who have never visited a physical store.

Question 4: Total revenue (selling price × quantity, not profit) for all stores in December 2020 - for the P&L board meeting.

Thinking Process

- The sales table does not have a price -> JOIN with products to get unit_price_usd.
- Clarify the definition of 'revenue' = unit_price_usd * quantity (without deducting the cost of goods sold).
- Filter order_date in the range of 01-31/12/2020, and calculate the total through a CTE to easily check each step.

SQL

```
-- Đi tính tổng doanh thu trên từng đơn hàng
WITH data_raw AS ( -- Tính số tiền chi tiết cho từng phân loại trên mỗi đơn hàng
SELECT
    s.order_number
    , sum(p.unit_price_usd * s.quantity) AS total_invoice_amount
FROM retails.sales s
INNER JOIN retails.products p
    ON s.product_key = p.product_key
WHERE 1 =1
    AND s.order_date BETWEEN '2020-12-01' AND '2020-12-31'
GROUP BY
    s.order_number
)
SELECT
    sum(total_invoice_amount) AS total_revenue_month12
FROM data_raw
```

Query Result

	total_revenue_month12
1	651526.44

Insight

- The revenue for December 2020 reached \$651,526, but compared to the same period in December 2019, which reached \$2.48 million, December 2020 decreased by 74%. Impacting the revenue factor involves many elements, which can come from within the company itself or external factors; this time period coincided with the outbreak of the COVID-19 pandemic, causing all economic activities to stall due to quarantine and lockdown orders.
- This is a warning figure needed in the board meeting: the peak season could no longer 'carry' the revenue for the entire year of 2020.

Question 5: How many stores does each country have? Sort descending to see which country dominates.

Thinking Process

- The online channel is saved as 1 'store' with state = 'Online' -> it must be excluded before counting physical stores.
- GROUP BY country, COUNT(*), and sort descending.

SQL

```

SELECT
    s.country
    , COUNT(*) AS total_store
FROM retails.stores s
WHERE s.state != 'Online'
GROUP BY s.country
ORDER BY total_store DESC

```

Query Result

	country	total_store
1	United States	24
2	Germany	9
3	United Kingdom	7
4	France	7
5	Australia	6
6	Canada	5
7	Netherlands	5
8	Italy	3

Insight

- The US dominates with 24 out of 66 stores (36%), nearly 3 times that of the second country, Germany (9). Italy has the fewest with only 3 stores.
- Combined with question 3 (Canada has many customers but only 5 stores) -> the store network and customer distribution are skewed; Canada is a strong candidate if opening more sales locations

Question 6: Top 5 best-selling products (cumulative quantity historically) in each category - serving the Q1 restock plan

Thinking Process

- CTE 1: Cumulate quantity by product_key over the entire sales history.
- CTE 2: JOIN with products to get category/name, use DENSE_RANK() PARTITION BY category to rank by decreasing sales volume (DENSE_RANK ensures tied products do not 'skip' ranks).
- Filter rank <= 5.

SQL

```

WITH data_raw AS (
    SELECT
        s.product_key
        , SUM(s.quantity) AS sum_quantity_product
    FROM retails.sales s
    GROUP BY s.product_key
),
ranked_products AS (
    SELECT
        p.category
        , p.product_name
        , d.sum_quantity_product
        , DENSE_RANK() OVER (PARTITION BY p.category ORDER BY d.sum_quantity_product DESC) AS
rank_quantity
    FROM retails.products p
    INNER JOIN data_raw d
        ON d.product_key = p.product_key
)
SELECT
    category
    , product_name
    , sum_quantity_product
    , rank_quantity
FROM ranked_products
WHERE rank_quantity <= 5
ORDER BY category, rank_quantity

```

Query Result

	category	product_name	sum_quantity_product	rank_quantity
1	Audio	WMI 1GB Digital Voice Recorder Pen E100 Black	431	1
2	Audio	WMI 1GB Digital Voice Recorder Pen E100 Red	388	2
3	Audio	WMI 4GB Video Recording Pen X200 Yellow	370	3
4	Audio	WMI 4GB Video Recording Pen X200 Black	366	4
5	Audio	WMI Wireless Bluetooth Stereo Headphones M270 Black	359	5
6	Cameras and camcorders	Contoso Cyber Shot Digital Cameras Adapter E306 Black	101	1
7	Cameras and camcorders	Contoso Digital Camera/Camcorder USB Cable E324 Purple	97	2
8	Cameras and camcorders	Contoso Macro Zoom Lens X300 Black	95	3
9	Cameras and camcorders	Contoso Carrying Case E312 Pink	95	3
10	Cameras and camcorders	Contoso General Soft Carrying Case E318 Black	93	4
11	Cameras and camcorders	Contoso General Soft Carrying Case E318 White	93	4
12	Cameras and camcorders	Contoso Digital Camera/Camcorder USB Cable E324 White	93	4
13	Cameras and camcorders	Contoso USB Cable M250 Black	92	5
14	Cell phones	The Phone Company Touch Screen Phones - LCD M12 Gold	256	1
15	Cell phones	Headphone Adapter for Contoso Phone E130 Black	229	2
16	Cell phones	The Phone Company Touch Screen Phones Infrared M901 Gold	224	3
17	Cell phones	Contoso Finger Touch Screen Phones M30 Black	223	4
18	Cell phones	The Phone Company Touch Screen Phones 5-Wire/On-wall M50...	217	5
19	Computers	WMI Desktop PC2.33 X2330 Black	550	1

Insight

- The 'best-selling' level varies greatly between categories: the top 1 in Computers sold 550 units, while the top 1 in Cameras only sold 101 -> we can see that the restock plan cannot use a common quantity threshold for all categories.
- However, in the Cameras and Camcorders Category, the entire top 5 are cheap accessories (USB cables, carrying cases, adapters) rather than actual cameras -> We can see that in this category, the products that are 'best-selling by quantity' do not 'contribute heavily to revenue'. We should add a column for revenue over the last 1 month, 3 months, 6 months, and 12 months before finalizing the inventory import list.

Question 7: Margin % = (selling price - cost price) / selling price, averaged by subcategory; only consider subcategories with >= 10 SKUs; sort descending

Thinking Process

- Calculate margin % at the product level: $(\text{unit_price} - \text{unit_cost}) * 100 / \text{unit_price}$.
- AVG by subcategory; use HAVING COUNT >= 10 to filter out groups with too few SKUs (the average is only statistically meaningful with enough data).

SQL

SELECT

p.subcategory

, ROUND(AVG((p.unit_price_usd - p.unit_cost_usd) * 100/p.unit_price_usd), 2) AS margin_product

FROM retails.products p

GROUP BY p.subcategory

HAVING COUNT(p.product_key) >= 10

ORDER BY margin_product DESC

Query Result

	subcategory ▼	margin_product ▼
1	Digital SLR Cameras	60.4400000000000
2	Digital Cameras	57.3900000000000
3	Projectors & Screens	57.3300000000000
4	Movie DVD	57.2400000000000
5	Monitors	56.7100000000000
6	Printers, Scanners & Fax	56.3800000000000
7	Camcorders	56.2600000000000
8	Smart phones & PDAs	55.9300000000000
9	Bluetooth Headphones	55.6900000000000
10	Touch Screen Phones	55.2600000000000
11	Televisions	55.2000000000000
12	MP4&MP3	55.0300000000000
13	Home & Office Phones	54.9100000000000
14	Recording Pen	54.8400000000000
15	Washers & Dryers	54.5900000000000
16	Water Heaters	54.5700000000000
17	VCD & DVD	54.4500000000000
18	Laptops	54.0700000000000
19	Air Conditioners	53.7800000000000

Insight

- The margin fluctuates in a fairly narrow band of 50.8% - 60.4%: the highest is Digital SLR Cameras (60.44%), and the lowest is Cell phones Accessories (50.8%). A difference of only about 10% indicates that the pricing policy is very consistent across product groups.

Question 8: For orders with delivery, what is the average number of days from ordering to receiving by the customer's country? Which country is the slowest?

Thinking Process

- Only home delivery orders have a delivery_date -> filter delivery_date IS NOT NULL (in-store purchases are NULL, and should not be counted as '0 days delivery').
- JOIN customers to get the customer's country; use DATEDIFF(DAY, order_date, delivery_date) then AVG by country.

SQL

SELECT

```
c.country
, COUNT(DISTINCT s.order_number) AS count_order
, AVG(DATEDIFF( DAY, s.order_date, s.delivery_date)) AS avg_delivery_date
FROM retails.sales s
INNER JOIN retails.customers c
    ON s.customer_key = c.customer_key
WHERE delivery_date IS NOT NULL
GROUP BY c.country
ORDER BY avg_delivery_date DESC
```

Query Result

	country ▼	count_order ▼	avg_delivery_date ▼
1	Australia	281	4
2	Canada	512	4
3	France	133	4
4	Germany	538	4
5	Italy	211	4
6	Netherlands	200	4
7	United Kingdom	637	4
8	United States	3068	4

Insight

- Italy is the slowest (4.77 days) and Australia is the fastest (4.32 days) -> but the difference across the entire board is only 0.45 days. Conclusion for stakeholders: NO country has 'issues' with SLA; the delivery system is uniform.
- Only about 21% of total orders incur delivery (the rest are purchased in-store) – the sample size for Italy/France is quite small (211/133 orders), so the top ranking may not be stable and does not provide an accurate and objective assessment of delivery time when compared to the United States/United Kingdom.

**Question 9: In each country, find the top 1 spending customer in 2020:
country, customer name, total spending.**

Thinking Process

- CTE 1: total spending in 2020 by customer_key (JOIN products to get price).
- CTE 2: JOIN customers to get country + name, use DENSE_RANK() PARTITION BY country descending by spending.

- Filter rank = 1 (DENSE_RANK ensures if 2 customers tie for first place, both are shown without omission).

SQL

```
WITH data_raw_total_spending AS( -- tính tổng doanh thu theo từng khách hàng
SELECT
    s.customer_key
    , SUM(s.quantity * p.unit_price_usd) AS total_spending
FROM retails.sales s
INNER JOIN retails.products p
    ON p.product_key = s.product_key
WHERE YEAR(s.order_date) = 2020
GROUP BY s.customer_key
),
data_raw_rank_country AS (
SELECT
    c.country
    , c.name
    , d.total_spending
    , DENSE_RANK() OVER ( PARTITION BY c.country ORDER BY d.total_spending DESC) AS
rank_spending_country
FROM data_raw_total_spending d
INNER JOIN retails.customers c
    ON c.customer_key = d.customer_key
)
SELECT
    country
    , name
    , total_spending
FROM data_raw_rank_country
WHERE rank_spending_country = 1
Query Result
```

	country	name	total_spending
1	Australia	Makayla Hassall	16778.46
2	Canada	Zane Belgrave	20644.89
3	France	Alice Lafond	18659.88
4	Germany	Daniel Kaestner	17931.00
5	Italy	Gaspere Trevisan	13831.89
6	Netherlands	Polat Klarenbeek	13369.52
7	United Kingdom	Dominic Banks	19371.06
8	United States	Mie Huus	33275.47

Insight

- The global #1 VIP is an American customer with \$33,275 in 2020 -> 2.5 times the VIP of the Netherlands (\$13,370). From this, we can see the spending level of each customer across different countries.
- For customers with high spending compared to the general level of the national market, we should have a separate and personalized care program that will help retain and increase their satisfaction with the company

Question 10: List all SKUs in the catalog that have never been sold - for the clearance team to handle.

Thinking Process

- The problem 'exists in A but not in B' -> use anti-join: LEFT JOIN sales then filter WHERE s.product_key IS NULL.

SQL

SELECT

```
p.product_key
, p.product_name
, p.brand
, p.category
```

FROM retails.products p

LEFT JOIN retails.sales s

```
ON s.product_key = p.product_key
```

WHERE s.product_key IS NULL

Query Result

	product_key	product_name	brand	category
1	274	Contoso Home Theater System 5.1 Channel M1530 White	Contoso	TV and Video
2	1885	Contoso Washer & Dryer 21in E210 Green	Contoso	Home Appliances
3	2190	Adventure Works Floor Lamp X1150 Black	Adventure Works	Home Appliances
4	2191	Adventure Works Floor Lamp M2150 Black	Adventure Works	Home Appliances
5	2193	Adventure Works Chandelier M6150 Black	Adventure Works	Home Appliances
6	2211	Adventure Works Wall Lamp E3150 Silver	Adventure Works	Home Appliances
7	2212	Adventure Works Desk Lamp E1300 Silver	Adventure Works	Home Appliances
8	2214	Adventure Works Floor Lamp X1150 Grey	Adventure Works	Home Appliances
9	2219	Adventure Works Wall Lamp E3150 Grey	Adventure Works	Home Appliances
10	2220	Adventure Works Desk Lamp E1300 Grey	Adventure Works	Home Appliances
11	2229	Adventure Works Desk Lamp E1200 Blue	Adventure Works	Home Appliances
12	2230	WMI Floor Lamp X115 Black	Wide World Importers	Home Appliances
13	2234	WMI Wall Lamp E215 Black	Wide World Importers	Home Appliances
14	2251	WMI Wall Lamp E315 Silver	Wide World Importers	Home Appliances
15	2274	Proseware Wall Lamp E0215 Black	Proseware	Home Appliances
16	2277	Proseware Desk Lamp E0120 Black	Proseware	Home Appliances
17	2280	Proseware Chandelier M0815 White	Proseware	Home Appliances
18	2282	Proseware Wall Lamp E0215 White	Proseware	Home Appliances
19	2295	Proseware Floor Lamp M0215 Grey	Proseware	Home Appliances

Insight

- Only 25 out of 2,517 SKUs (about 1%) have never been sold -> a very low rate, meaning the catalog is generally 'healthy'.
- But the distribution is highly concentrated: 24 out of 25 zombie SKUs belong to Home Appliances, and 9 out of 25 belong to the Adventure Works brand. This is not a random phenomenon. From there, the merchandising team should review the shelf-listing and pricing specifically for this home appliance group instead of doing a broad clearance.

Question 11: Monthly revenue and cumulative revenue over the last 24 months - for the CEO's slide: monthly to see the trend, cumulative to see the scale.

Thinking Process

- Anchor the dynamic timeframe by MAX(order_date) in the data (do not hard-code dates) then go back 24 months using DATEADD.
- Group revenue by month (FORMAT yyyy-MM for display, DATEFROMPARTS to ensure proper chronological sorting).
- Calculate the cumulative total using the SUM() OVER (ORDER BY month) window function on the grouped result.

SQL

```
WITH data_raw_1 AS (  
    SELECT  
        FORMAT(s.order_date, 'yyyy - MM') AS YearMonth  
        , DATEFROMPARTS( YEAR(s.order_date), MONTH(s.order_date), 1) AS MonthStart  
        , SUM( s.quantity * p.unit_price_usd) AS total_revenue_bill  
    FROM retails.sales s  
    INNER JOIN retails.products p  
        ON p.product_key = s.product_key  
    WHERE s.order_date >= DATEADD( MONTH, -24, (SELECT MAX(order_date) FROM retails.sales))  
    GROUP BY FORMAT( s.order_date, 'yyyy - MM'), DATEFROMPARTS( YEAR(s.order_date),  
MONTH(s.order_date), 1)  
)  
SELECT  
    YearMonth  
    , total_revenue_bill  
    , SUM(total_revenue_bill) OVER( ORDER BY MonthStart) AS cumulative_revenue  
FROM data_raw_1 d1  
ORDER BY MonthStart
```

Query Result

	YearMonth ▾	total_revenue_bill ▾	cumulative_revenue ▾
1	2019 - 02	829119.85	829119.85
2	2019 - 03	845925.09	1675044.94
3	2019 - 04	149892.71	1824937.65
4	2019 - 05	1594446.47	3419384.12
5	2019 - 06	1404861.40	4824245.52
6	2019 - 07	1408714.62	6232960.14
7	2019 - 08	1500784.77	7733744.91
8	2019 - 09	1547870.73	9281615.64
9	2019 - 10	1575168.82	10856784.46
10	2019 - 11	1715659.88	12572444.34
11	2019 - 12	2477295.85	15049740.19
12	2020 - 01	2068951.12	17118691.31
13	2020 - 02	2227379.57	19346070.88
14	2020 - 03	674990.66	20021061.54
15	2020 - 04	217642.10	20238703.64
16	2020 - 05	878026.63	21116730.27
17	2020 - 06	763883.23	21880613.50
18	2020 - 07	512805.18	22393418.68
19	2020 - 08	416647.08	22810065.76

Insight

- Total revenue for the last 24 months reached \$25.4 million, but the trend line tells a clear story: peaking at \$2.48M in 12/2019, maintaining >\$2M until 02/2020, then free-falling from 03/2020 – hitting a bottom of \$218k in 04/2020 (-90% compared to the peak), coinciding with the outbreak of COVID-19.

Question 12: What percentage of first-time buyers in year X return in year X+1, X+2, X+3? Group customers by their first purchase year and export a retention matrix

Thinking Process

1. CTE 1: xác định năm mua ĐẦU TIÊN của mỗi khách (MIN order_date) — đây là 'cohort'.
2. CTE 2: với mỗi khách, liệt kê các năm có mua hàng và tính offset = năm mua – năm cohort.
3. CTE 3: đếm DISTINCT khách theo (cohort, offset); cohort size = số khách tại offset 0.

4. Pivot bằng MAX(CASE WHEN offset = 1/2/3 ...) chia cho cohort size ra % retention.

SQL

```
WITH _year_purchase AS (  
  SELECT  
    DISTINCT s.customer_key  
    , YEAR(MIN(s.order_date)) AS year_start_buy  
  FROM retails.sales s  
  GROUP BY s.customer_key  
) , other_year AS (  
  SELECT  
    DISTINCT yp.customer_key  
    , yp.year_start_buy  
    , YEAR(s.order_date) AS other_year  
    , YEAR(s.order_date) - yp.year_start_buy AS number_client_year  
  FROM _year_purchase yp  
  INNER JOIN retails.sales s  
    ON s.customer_key = yp.customer_key  
) , cohort_analysis AS (  
  SELECT  
    ot.year_start_buy  
    , ot.number_client_year  
    , COUNT( DISTINCT ot.customer_key) AS count_customer  
  FROM other_year ot  
  GROUP BY ot.year_start_buy, ot.number_client_year  
) , cohort_size AS (  
  SELECT  
    ca.year_start_buy  
    , ca.count_customer AS total_customer  
  FROM cohort_analysis ca  
  WHERE ca.number_client_year = 0  
)  
  
SELECT  
  cs.year_start_buy  
  , ROUND( 100.0 * MAX( CASE WHEN ca.number_client_year = 1 THEN ca.count_customer END) /  
cs.total_customer, 2) AS retention_1yr  
  , ROUND( 100.0 * MAX( CASE WHEN ca.number_client_year = 2 THEN ca.count_customer END) /  
cs.total_customer, 2) AS retention_2yr  
  , ROUND( 100.0 * MAX( CASE WHEN ca.number_client_year = 3 THEN ca.count_customer END) /  
cs.total_customer, 2) AS retention_3yr  
FROM cohort_size cs  
INNER JOIN cohort_analysis ca  
  ON cs.year_start_buy = ca.year_start_buy  
GROUP BY cs.year_start_buy, cs.total_customer  
ORDER BY cs.year_start_buy
```

Query Result

	year_start_buy ▾	retention_1yr ▾	retention_2yr ▾	retention_3yr ▾
1	2016	20.730000000000	32.760000000000	45.650000000000
2	2017	34.600000000000	46.510000000000	27.060000000000
3	2018	44.810000000000	27.290000000000	3.640000000000
4	2019	25.420000000000	2.860000000000	NULL
5	2020	3.590000000000	NULL	NULL
6	2021	NULL	NULL	NULL

Insight

- Year-1 retention improved continuously from the 2016 cohort (20.7%) to a peak in the 2018 cohort (44.8%) -> during this period, customer retention was gradually improving between 2016 and 2018.
- Then it reversed: the 2019 cohort dropped to 25.4% and the 2020 cohort fell to 3.6% -> meaning almost all new customers in 2020 did not return. We can see that retention started to decrease from 2019 (down 19.4%) regarding return visits after 1 year of purchase. The cause could be due to quarantine measures when the COVID-19 pandemic broke out, making ordering and delivery quite difficult. It does not stem from the company's inability to adapt quickly, because revenue from direct physical stores only accounts for about 20.88% of total revenue.

Question 13: Revenue per square meter of each store to compare space utilization efficiency, sorted into 4 quartile groups within the same country.

Thinking Process

- CTE 1: total revenue by store in the time window; CTE 2: divide by square_meters, excluding the Online channel (which has no physical area).
- Calculate NTILE(4) ONCE in a separate CTE and then assign labels using CASE -> if NTILE is called repeatedly in each WHEN branch, every call acts as a separate window calculation, which is prone to errors and hard to read

SQL

```
WITH total_revenue AS (
  SELECT
    s.store_key
    , sum(s.quantity * p.unit_price_usd) AS order_value
  FROM retails.sales s
  INNER JOIN retails.products p
    ON s.product_key = p.product_key
```

```

WHERE s.order_date >= DATEADD(MONTH, -12, (SELECT MAX(order_date) FROM retails.sales)) -- 12 tháng
gan nhat: du lieu ket thuc 02/2021 nen YTD 2023 se rong
GROUP BY s.store_key
), revenue_acreage AS (
SELECT
    st.store_key
    , st.country
    , tr.order_value / st.square_meters AS square_revenue
FROM retails.stores st
INNER JOIN total_revenue tr
    ON tr.store_key = st.store_key
WHERE st.country != 'Online'
), quartile_rank AS (
SELECT
    ra.store_key
    , ra.country
    , ra.square_revenue
    , NTILE(4) OVER( PARTITION BY ra.country ORDER BY ra.square_revenue DESC) AS quartile
FROM revenue_acreage ra
)
SELECT
    qr.store_key
    , qr.country
    , qr.square_revenue
    , CASE qr.quartile
        WHEN 1 THEN 'Top 25%'
        WHEN 2 THEN 'Upper-middle'
        WHEN 3 THEN 'Lower-middle'
        ELSE 'Bottom 25%'
    END AS quartile_label
FROM quartile_rank qr
ORDER BY qr.store_key

```

Query Result

	store_key ▾	country ▾	square_revenue ▾	quartile_label ▾
1	1	Australia	67.815915	Top 25%
2	4	Australia	9.937020	Bottom 25%
3	5	Australia	43.946805	Upper-middle
4	6	Australia	35.850345	Lower-middle
5	8	Canada	77.130251	Lower-middle
6	9	Canada	110.366920	Upper-middle
7	10	Canada	120.435685	Top 25%
8	12	France	111.515114	Top 25%
9	13	France	113.652408	Top 25%
10	14	France	20.235657	Bottom 25%
11	15	France	74.359275	Upper-middle
12	16	France	61.085584	Lower-middle
13	17	France	83.424057	Upper-middle
14	18	France	63.122612	Lower-middle
15	19	Germany	51.443559	Upper-middle
16	21	Germany	92.668178	Top 25%
17	22	Germany	22.585425	Bottom 25%
18	23	Germany	42.785736	Lower-middle
19	24	Germany	49.666345	Upper-middle

Insight

- Area efficiency varies by up to 20 times: the best store is Saarland, Germany - only 350m² but reached \$199.6/m²/year; the worst is Tasmania, Australia - 2,000m² but only reached \$9.9/m².
- Common pattern: small stores (350-1,300m²) dominate the top efficiency spots, while the 2,000m² 'hypermarkets' cluster at the bottom of the table -> indicating that the small store format is winning; this is worth considering when deciding on new premises.

Question 14: Which pairs of products often appear together in 1 order? Top 20 pairs, only considering orders with >= 2 different products - input for the recommendation engine

Thinking Process

- CTE 1: filter valid orders with >= 2 different products (HAVING COUNT(DISTINCT product_key) > 1).

- Self-join the sales table with itself by order_number; condition product_key_A < product_key_B so each pair is counted only once (avoiding both (A,B) and (B,A), and eliminating self-pairing (A,A)).
- Count the number of orders containing each pair, divide by the total number of valid orders to get the % rate; take TOP 20.

SQL

```
WITH valid_orders AS (
  SELECT
    DISTINCT s.order_number
    , COUNT( DISTINCT s.product_key) AS count_item
  FROM retails.sales s
  GROUP BY s.order_number
  HAVING COUNT( DISTINCT s.product_key) > 1
), product_pairs AS (
  SELECT
    vo.order_number
    , s1.product_key AS product_1
    , s2.product_key AS product_2
  FROM retails.sales s1
  JOIN retails.sales s2
    ON s1.order_number = s2.order_number
  INNER JOIN valid_orders vo
    ON vo.order_number = s1.order_number
  WHERE s1.product_key < s2.product_key
), count_pair_product AS (
  SELECT
    pp.product_1
    , pp.product_2
    , COUNT( DISTINCT pp.order_number) AS pair_count
    , (
      SELECT
        COUNT( DISTINCT vo.order_number)
      FROM valid_orders vo
    ) AS count_all
  FROM product_pairs pp
  GROUP BY pp.product_1, pp.product_2
)
SELECT
  TOP 20
  cpp.product_1
  , cpp.product_2
  , cpp.pair_count
  , cpp.pair_count * 100.0 / cpp.count_all AS rate_pair
FROM count_pair_product cpp
ORDER BY pair_count DESC
```

Query Result

	product_1 ▾	product_2 ▾	pair_count ▾	rate_pair ▾
1	1648	1699	5	0.029203901641
2	101	417	4	0.023363121313
3	1630	2503	4	0.023363121313
4	421	434	4	0.023363121313
5	46	416	4	0.023363121313
6	430	435	4	0.023363121313
7	417	424	4	0.023363121313
8	432	440	4	0.023363121313
9	75	448	4	0.023363121313
10	97	444	4	0.023363121313
11	447	450	4	0.023363121313
12	450	455	4	0.023363121313
13	92	459	4	0.023363121313
14	438	1572	4	0.023363121313
15	61	1573	4	0.023363121313
16	460	1590	4	0.023363121313
17	434	1581	4	0.023363121313
18	60	1598	4	0.023363121313
19	421	1597	4	0.023363121313

Insight

- 65% of orders (17,121 / 26,326) have 2 or more products -> meaning customers very often buy multiple items, seemingly a gold mine for recommendations.
- But the result is surprising: the most popular pair only appeared 5 times (about 0.03% of orders). With 2,500 SKUs, the combinations are so dispersed that NO product pair is strong enough to create an 'if you buy A, you buy B' type of suggestion rule.